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Curative effect of arjunolic acid from *Terminalia arjuna* in non-alcoholic fatty liver disease models

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ABSTRACT

The prevalence of Non Alcoholic Fatty Liver Disease (NAFLD) is increasing globally. *Terminalia arjuna* W. & Arn. (Combretaceae) is an endemic tree found in India and Sri Lanka and used traditionally for its cardioprotective and hepatoprotective effects. Arjunolic acid (AA) is an oleanane triterpenoid found mainly in the heartwood of *T. arjuna.* This study was aimed to evaluate the hepatoprotective effect of AA using cellular and rodent models of NAFLD. AA was isolated from the ethyl acetate extract of the heartwood of *T. arjuna.* The structure of AA was confirmed by physical and spectroscopic data. Steatosis was induced in HepG2 cells using palmitate-oleate mixture and the effects of AA on triglyceride accumulation and lipotoxicity were assessed. *In vivo* effect of AA on NAFLD was assessed using HFD fed rats. The treatment with AA did not affect the cell viability upto 100 μM and showed GI_{25} value of 379.9 μM in HepG2 cells. The treatment with AA significantly lowered the ORO concentration by 35.98% and triglyceride accumulation by 66.36% at 50 μM concentration ($P < 0.005$) compared to the vehicle treated group. The treatment with AA also reduced the leakage of ALT and AST by 61.11 and 48.29% in a significant manner ($P < 0.005$). The *in vivo* findings clearly demonstrated that the animals treated with AA at 25 and 50 mg/kg concentrations showed a significant decrease in the levels of transaminases, phosphatase and GGT ($P < 0.005$). In the liver, the expression of PPARα and FXRα expressions were upregulated, while PPARγ expression was downregulated by the treatment with AA. The liver histology of the animals showed reduction in steatosis and MNC infiltration. These preliminary evidences suggested that AA might be a promising lead to treat NAFLD. Future robust scientific studies on AA will lead to tailoring it for the treatment of NAFLD.

1. Introduction

Non-alcoholic fatty liver disease (NAFLD) is one of the chronic liver diseases its identified as one of the potential risk factors for hepatic and cardiometabolic mortality. NAFLD is defined as excessive fat accumulation in the liver, as demonstrated by imaging or histology without a history of significant alcohol consumption. The pathological spectra of NAFLD range from simple steatosis to steatohepatitis with progression towards fibrosis, cirrhosis and hepatocellular carcinoma [1,2]. Pathophysiology of NAFLD remains unclear; it is believed that the liver insults such as insulin resistance and metabolic syndromes lead to NAFLD [3]. Insulin resistance increases free fatty acids in serum which aggravate lipogenesis in the liver leading to hepatic steatosis [4]. Studies showed that 70–80% of individuals with NAFLD are predominantly found to be obese and/or T2D [5,6]. Present studies reveal that the global prevalence of obesity will hit 18% in men and over 21% in

Abbreviations: bw, body weight; CMC, carboxymethylcellulose; CVD, cardiovascular disease; DMEM, Dulbecco's Minimal Essential Medium; DMSO d_6, deuterated dimethyl sulfoxide; FBS, fetal bovine serum; GI, growth inhibition concentration; HFD, high fat diet; KBr, potassium bromide; lit. mp., melting point given in the literature; MNC, mono-nuclear cells; mp, melting point; NFD, normal fat diet; ORO, oil red - O; PO-BSA, palmitate, oleate, BSA complex; SD, standard deviation; T2D, type 2 diabetes; Tc, total cholesterol; Tg, triglyceride

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women in 2025 [7]. The prevalence of NAFLD was high in various parts of the globe particularly in the Middle East (32%), South America (31%), Asia (27%), USA (24%), Europe (23%) and least common in Africa [8]. Lifestyle modification with dietary changes, regular exercise and cognitive behavior therapy are some of the effective therapeutic options [3]. Various pharmacological interventions are being tried in both research and clinical settings [9]; however there is no definite pharmacotherapy available for the treatment of NAFLD.

Terminalia arjuna (Roxb.) W. & Arn. (Family: Combretaceae) is commonly used in various traditional systems of medicine in India such as *ayurveda*, *siddha* and *unani* [10]. Various triterpenoids such as arjunin, arjunic acid, arjungenin, terminic acid, terminoltin and their glucosides, phenolic compounds such as arjunone, luteolin, baicalein, kampferol, pelargonidin and pyrocatechols were reported from *Terminalia arjuna.* Arjunolic acid (2α, 3β, 23-trihydroxy- olean-12-en-28-oic acid) is one of the major oleanane type, pentacyclic triterpenoids isolated from the heartwood of *Terminalia arjuna* [11]. Naturally, AA was 1.32 nm long and the usefulness of AA and its derivatives for the preparation of functional molecular frameworks in nanomaterials was reported [12]. Oleanane type is one of the majorly occurring classes of pentacyclic triterpenoids and their usefulness on the management of diabetes and its complications was reported [13]. The hepatoprotective, cardioprotective, renoprotective, antidiabetic, antioxidant, anticancer effects of AA were reviewed by Ghosh and Sil [14]. Hepatoprotective effect of AA was already reported in arsenic [15], acetaminophen [16] and atorvastatin [17] induced liver damage in laboratory animals. Lin and his coworkers [18] reported the hepatoprotective efficacy of chloroform extract of *Cyclocarya paliurus* in HFD fed animals, which was shown to have AA as one of its components. In this study, we evaluated the curative efficacy of AA in PO-BSA induced steatotic HepG2 cells and HFD fed rats.

2. Methodology

2.1. Chemicals

DMEM, FBS and Trypsin were purchased from Gibco (India); DMSO-d_6, Fenofibrate, HEPES, MTT, ORO, sodium palmitate and sodium oleate were purchased from Sigma Aldrich (Bangalore, India). Antibiotic-antimycotic mixture, gentamycin and LDH assay kit were obtained from Hi-Media chemicals (Mumbai, India). RNAlater, RNeasy mini kit and QuantiTect Reverse Transcription Kit were obtained from Thermo Scientific (Germany). Kits used for biochemical assessment were purchased from Accurex biomedical Pvt Ltd (Mumbai, India) and Agappe diagnostics (Kochi, India). All other chemicals and solvents of analytical grade were purchased from Qualigens.

2.2. Isolation of AA

2.2.1. Instrumentation

Melting point was taken on a heating block apparatus. IR spectrum was taken on Perkin-Elmer FT-IR grating spectrophotometer (Spectrum Two) in KBr disc. ^{1}H NMR and ^{13}C NMR spectra were recorded on Bruker AV-500 in DMSO-d_6 at 500 and 125 MHz, respectively. Chemical shifts were recorded as δ scale with TMS as the internal standard. MS was taken on a Jeol GC–MS (GC-Mate II) instrument.

2.2.2. Plant material

Heartwood of *T. arjuna* was collected from Tirunelveli District, Tamilnadu, India and its botanical identity was authenticated by Dr V. Chelladurai, Survey of Medicinal Plants Unit, Central council of Research in *Siddha*, Palayamkottai. A voucher specimen (ERI-EP-ET-05) was deposited in the herbarium of Entomology Research Institute, Loyola College, Chennai.

2.2.3. Extraction and purification

Coarsely powdered heartwood of *T. arjuna* (1.0 kg) was exhaustively extracted by refluxing with ethylacetate (5 L) in a bolt head round bottom flask fitted with a reflux condenser for five hours. Ethylacetate was selected as the suitable solvent for extraction on the basis of previous work [11]. The solvent was filtered while hot using Whatman No. 1 filter paper, under reduced pressure. The filtrate was concentrated in a rotary evaporator to about 500 mL, stored at 4 °C for 48 h and the precipitated crude AA (25 g) was collected by filtration. Crude AA was then purified by flash chromatography on silica gel (Avra, 400 mesh) in chloroform and eluted with the mixture of chloroform: methanol (19:1). Pure AA was obtained by repeated crystallization from methanol (mp. 334–335; lit.mp 337; Yield 16 g).

2.3. In silico studies

2.3.1. Drug score analysis

The partition coefficient (c log P) and (c log S) data regarding hydrogen bond acceptors and donors counts, bioavailability score, mutagenic, tumorigenic, reproductive effect, irritant and drug likeness scores of AA are predicted using DataWarrior software (Version 4.5.2) [19]. Based on the above values drug score was also calculated using the same software.

2.3.2. Molecular docking analysis

For this analysis, Autodock tools v1.5 and Autodock v4.2 programme from the Scripps Research Institute (http://www.scripps.edu/mb/olson/doc/autodock), PyMOL 1.3 Molecular Graphics System, and Chem Draw Ultra 11 installed on a desktop equipped with Intel(R) Core (TM) at 2.30 GHz, 2.40 GHz processor (4GB RAM Core CPU) running the Ubuntu 14.04 (LINUX) and Windows 7 operating system were used. ChemDraw Ultra, 11.0 was used to generate the three dimensional structure of ligands. PRODRG server [20] was used for the structural conformation and energy minimization of the ligand. The structure of PPARα (PDB ID: 1I7G) was recovered from Protein Data Bank (PDB) webpage and the CASTp was used to explore the possible binding sites of doki favored target receptors [21]. The ligands were docked to target protein complex PPARα using Auto dock tools [22,23] in accordance with our previous work [24,25]. The electrostatic map and affinity maps for all the atoms present were computed with a grid spacing of 0.375 Å. The Lamarckian Genetic Algorithm was used for docking search with a population of 150 individuals with a mutation rate of 0.02 evolved for 10 generations. The results were evaluated by sorting the different orientations of the ligands with respect to ligand-PPARα interactions, which were developed by Pose View [26], PyMol molecular viewer (The PyMOL Molecular Graphics System, Version 1.5.0.4 Schrodinger, LLC).

2.4. Evaluation of in vitro anti-NAFLD effect of AA on HepG2 cell lines

2.4.1. Cell culture

The human hepatocellular cell line (HepG2 cells) was obtained from National Center for Cell Sciences, Pune (India) and cultured in DMEM supplemented with heat inactivated FBS 10% (v/v), penicillin (100 U/mL), streptomycin (100 µg/mL) and gentamycin (100 µg/mL). The cells were maintained in a humidified incubator with 5% CO_2 at 37 °C. The cells were subcultured at 80% confluence by total media replacement using 0.25% (w/v) trypsin – 0.53 mM EDTA. DMSO at 0.01% in DMEM was used as the vehicle for all *in vitro* bioassays.

2.4.2. Effect of AA on the viability of HepG2 cells

Cytotoxic effect of AA on HepG2 cells was investigated using MTT assay [27]. Briefly, the cells were seeded in 96-well plate at a concentration of 1×10^4 cells/well. After 24 h, the spent media were replaced with fresh media containing test materials. The cytotoxicity of AA was assessed between 12.5–400 µM concentrations. Vehicle control cells were treated with DMSO alone. After 24 h of treatment, the cells were treated with 100 μL/well of sterile MTT (5 mg/mL in PBS; pH-7.4) and were incubated at 37 °C for 3 h. Then, the formazan crystals were solubilized in 200 μL/well of DMSO and the absorbance was measured at 492 nm using a Bio-Rad iMark™ microplate reader. The results of three independent experiments were taken for analysis. GI_{25} value of AA was calculated using linear regression analysis using GraphPad Prism (Version 5.0) using percent cell viability value. Percentage cell viability was calculated using the following formula:

$$Percent\ cell \quad viability = \frac{A492 \quad nm\ (test\ material)}{A492 \quad nm\ (control)} \times 100$$

2.4.3. Effect of AA on PO-BSA induced steatotic HepG2 cells

Steatosis induction was done using the protocol of Hetherington and coworkers [28] with a minor modification [29,30]. HepG2 cells were seeded in 24 well plates at a density of 4×10^4 cells/well and incubated overnight. On the following day, the spent media were replaced with the fresh media having PO-BSA with or without test materials. The range for AA for this study was fixed on the basis of MTT assay, between 12.5 and 50 μM. Fenofibrate was used as positive control and its dose was fixed as 50 μM [30]. The cells replenished with media containing only BSA without PO-BSA but treated with DMSO served as the normal control. The cells treated with DMSO in media containing PO-BSA served as negative control. The treated cells were incubated overnight in a humidified atmosphere with 5% CO_2 at 37 °C.

2.4.3.1. ORO staining and Tg estimation. The level of lipid accumulation was measured by ORO staining [31]. After 24 h of incubation, spent medium was removed and the cells were fixed with 10% buffered formalin in PBS (pH-7.4). The wells were then washed with 60% isopropanol and then incubated with 60% ORO solution for 10 min. After washing with deionized water, the morphology of the cells was evaluated and captured (400×) using an inverted microscope. The ORO was eluted in 100% isopropanol and the extracted dye was quantified by measuring the absorbance at 520 nm. The results were given as mean percent absorption of the sample to control for three independent experiments.

After treatment, the cells were harvested and lysed using lysis buffer having 1% Triton X-100 in PBS. The levels of Tg and total protein in the cell lysate were assessed by GPO-PAP and Lowry's method, respectively using commercial kits. The level of Tg in the cell lysate was normalized with protein level and the results were given as mg Tg per g protein for three independent experiments.

2.4.3.2. Estimation of LDH, ALT and AST. Effect of AA on defense of hepatic cells from lipotoxicity stimulated by PO-BSA was measured by assessing the LDH and transaminase levels. After 24 h treatment, the phenolic red free spent media were collected in sterile tubes. The cells were rinsed with sterile PBS and lysed with lysis buffer. The enzyme levels were assessed in the supernatant. The data were given as the mean of three independent experiments.

2.5. Effect of AA in HFD fed animals

2.5.1. Animals

Male Wistar rats (160–175 g) obtained from the King Institute of Preventive Medicine, Chennai were used in this study. The rats were acclimatized in polypropylene cages at 23 ± 1 °C with a relative humidity of 50 ± 5% and 12/12 light/dark cycles. The animals were given water and feed *ad libitum.* This study was carried out in accordance with the guidelines of CPCSEA and approved by the Institutional Animal Ethics Committee (IAEC-ERI-LC-01/16).

2.5.2. Dose fixation and preparation of test materials

Previous studies in rats with AA established the dose range between 100 and 200 mg/kg [32,33] and for this study the dose level was fixed as 25 and 50 mg/kg bw. Fernofibrate (50 mg/kg bw) was used as positive control and the vehicle alone served as negative control. The test materials were suspended in a vehicle 0.5% CMC, 0.9% NaCl, 0.3% Tween-80 and 98.1% distilled water. The treatment was administered once daily between 3.00–4.00 p.m., through oral gavage for four weeks.

2.5.3. Evaluation of acute toxic effect of AA in rats

Acute toxicity was evaluated according to OECD guideline #423 [34] without any modification. The animals were acclimatized for five days before the start of the experiment and 300 mg/kg was fixed as the starting dose for the experiment. Three animals were dosed with AA at 300 mg/kg concentration and observed individually for toxic reactions, onset and recovery periodically for 14 days. Since there was no observable toxicity at this dose level, further three animals were dosed with AA at 2000 mg/kg concentration. To confirm the findings, an additional three animals were dosed with AA at 2000 mg/kg concentration, as per the guidelines.

2.5.4. Experimental design

The animals were fed either with NFD or HFD for eight weeks before the start of the experiment and throughout experimental period in accordance with our previously published methodology [30]. Based on the body weight, the animals were randomly grouped into five having six animals in each group. Group I consisted of animals fed NFD and treated with vehicle (normal control); Group II consisted of HFD fed animals treated with vehicle (negative control); Group III consisted of HFD fed animals treated with fenofibrate at 50 mg/kg concentration (positive control); and Groups IV and V consisted of HFD fed animals treated with AA at 25 and 50 mg/kg concentrations, respectively. The treatment was given for four weeks through oral gavage.

2.5.5. End of the study

On completion of treatment for four weeks, the animals were euthanized under CO_2 anesthesia and blood was collected through cardiac puncture and stored in EDTA coated tubes. Plasma was separated from blood after centrifugation at 3500 rpm for 15 min and stored at 4 °C for biochemical assays. Liver tissues were rapidly harvested and stored in RNase Later for RNA extraction and in 10% buffered formalin with PBS for histopathological analysis, respectively.

2.5.6. Parameters

AST (Accurex, #L06217), ALT (Accurex, #L04517), ALP (Accurex, #15417), LDH (Accurex, #L11018), GGT (Accurex, #03317), Tc (Agappe, #11403002), Tg (Agappe, #114140002), HDL#L06217), ALT (Accurex, #L04517), ALP (Accurex, #15417), LDH (Accurex, #L11018), GGT (Accurex, #03317), Tc (Agappe, #11403002), Tg (Agappe, #114140002), HDL-c (Agappe, #11414004), LDL-c (Agappe, #11414007) and levels were estimated on the same day of euthanasia using commercial spectrophotometric assay kits.

2.5.7. Assessment of mRNA expression levels on HFD fed rats treated with AA

Primers of genes such as PPARα, PPARγ, FXR and β-actin were designed in consultation with previously published literature [35,36]. Total RNA was extracted from liver tissue using RNeasy kit; it was quantified using Nanodrop and subjected to RT-PCR reaction using QuantiTect Reverse Transcription kit. The RT-PCR reaction was on the final volume of 20 μL. Genomic DNA elimination reaction containing gDNA wipeout buffer, template RNA and RNase-free water for 14 μL was incubated at 42 °C for 2 min followed by addition of 6 μL reverse-transcription reaction master mix and further incubated at 42 °C for 30 min with final heating for 3 min at 95 °C to inactivate Quantiscript Reverse Transcriptase. Then 35 cycles of PCR were performed on the following conditions: initial denaturation at 95 °C for 5 min, annealing denaturation at 95 °C for 1 min, annealing temperature depending on

Table 1
Computational estimation of pharmacokinetics and toxicity of AA.

Properties	AA
c log P	4.7215
c log S	−5.663
H bond acceptor	5
H bond donor	4
Bioavailability score	0.56
Mutagenic	None
Tumorigenic	None
Reproductive effects	None
Irritant	None
Drug score	0.250

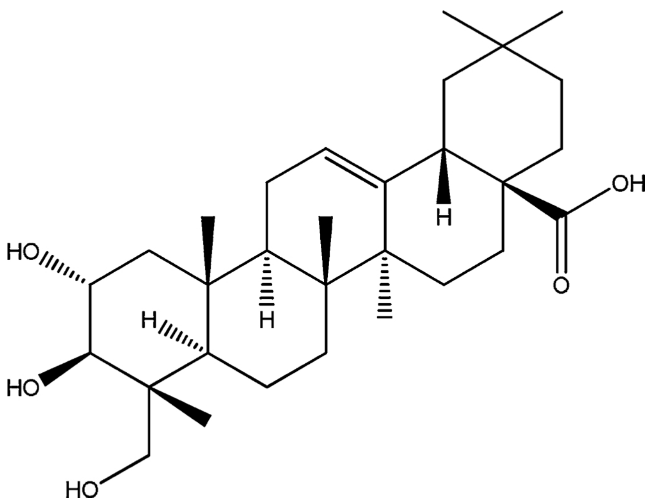

Fig. 1. Structure of AA.

individual primers for 1 min, and extension temperature of 72 °C for 1 min followed by a final extension step at 72 °C for 5 min. Primer sequences are given as supplementary Table 1. The PCR products were separated in 10% PAGE and visualized by ethidium bromide staining (0.02%). Transcripts were normalized on arbitrary density units corresponding to β-Actin band using Image J software.

2.6. Statistics

All the data were presented as mean ± SD. Homogeneity of variances in the data was assessed using Bartlett's test. If the data were homogeneous, statistical significance was assessed by one way ANOVA followed by Tukey's HSD. If the data were non-homogeneous, the statistical significance was assessed using Kruskal–Wallis test followed by Dunn's multiple comparison test (GraphPad Prism, version 5.0). The $P \leq 0.05$ was set as statistical significance for all the analyzed data.

3. Results

3.1. Phytochemistry

AA (Fig. 1) answered the Nollers' test for triterpenoid (pink colour with tin and thionyl chloride). The identity of the compound was confirmed by comparing the physical and spectroscopic data of the compound with those reported for AA [37]. The identity was further confirmed by direct comparison with an authentic sample.

3.2. In silico analysis

In silico investigations of *c* log *P*, *c* log *S*, H bond acceptors, H bond donors and bioavailability scores of AA are given in Table 1. AA did not show any mutagenic, tumorigenic and irritant properties. The binding profile of the AA on PPARα showed (Table 2) that the β equatorial OH at C-3 showed interaction with TYR-334/HN and the α equatorial OH at C-2 showed binding with THR-279/HG1. The COOH at C-7 showed binding with HIS-274/HE2 (Fig. 2a). Hydrophobic interactions of AA with active sites of PPARα are displayed in figure (Fig. 2b). The binding profile of the fenofibrate with PPARα (1I7G) showed that the ester carbonyl of the side chain bound with ASN-219/2HD2 and the ethereal oxygen connecting the benzene ring and the gem dimethyl carrying carbon bound with TYR-334/HN (Fig. 3a). Hydrophobic interactions of fenofibrate with active sites of PPARα are displayed in figure (Fig. 3b).

3.3. Effect of AA on the viability of HepG2 cell lines

Effect of AA on HepG2 cells viability was analyzed using MTT assay. The treatment of AA did not show toxicity up to 100 μM. Towards HepG2 cells, AA showed 746.34 and 379.9 μM as GI_{50} and GI_{25} values, respectively. Based on these data, the dose range of AA for *in vitro* studies was fixed between 12.5 and 50 μM, which were below to GI_{25} value of AA (Fig. 4).

3.4. Effect of AA treatment on lipid accumulation of PO-BSA induced steatosis in HepG2 cells

Incubation of HepG2 cells with PO-BSA complex caused rapid intracellular aggregation of lipids by about 60% compared to the normal cells. The treatment with AA showed significant reduction in lipid aggregation as indicated by the reductions in ORO concentrations of 24.89 ($P < 0.005$), 21.44 ($P < 0.05$) and 35.98% ($P < 0.005$) at 12.5, 25 and 50 μM concentrations, respectively. The treatment also significantly reduced ($P < 0.005$) the levels of triglyceride by 55.67, 62.56 and 66.36% at 12.5, 25 and 50 μM concentrations, respectively. Fenofibrate showed 26.14 and 59.69% reductions in the ORO concentration and triglyceride level in a significant manner ($P < 0.005$) (Table 3). Representative microphotographs of untreated and treated cells and stained with ORO also clearly showed positive effect of AA on lipid accumulation in hepatocytes (Fig. 5). Effect of AA on PO-BSA induced cytotoxicity was assessed by measuring the transaminase leakage in the spent media after 24 h of treatment. The treatment reduced the leakage of ALT and AST levels by 61.11 and 48.29% in a significant manner ($P < 0.005$) (Table 3).

3.5. Toxicity effected of AA on the Normal animals

The treatment with AA had not shown any noticeable adverse effect on rats up to 2 g/kg concentration; LD_{50} of AA should be > 2 g/kg and it fell in the category 5, as per the OECD guideline #423.

Table 2
Docking AA and fenofibrate with PPARα protein (PDB ID: 1I7G).

Ligand	Binding amino acid Residues	Binding Energy (kcal/mol)	Inhibition Constant μM	VDW HB dissolve energy (kcal/mol)	Ligand efficiency
AA	THR`279/HG1, HIS`274/HE2, TYR`334/HN	−6.62	14.03	−7.89	0.18
Fenofibrate	ASN`219/2HD2, TYR`334/HN	−7.24	4.93	−9.28	0.29

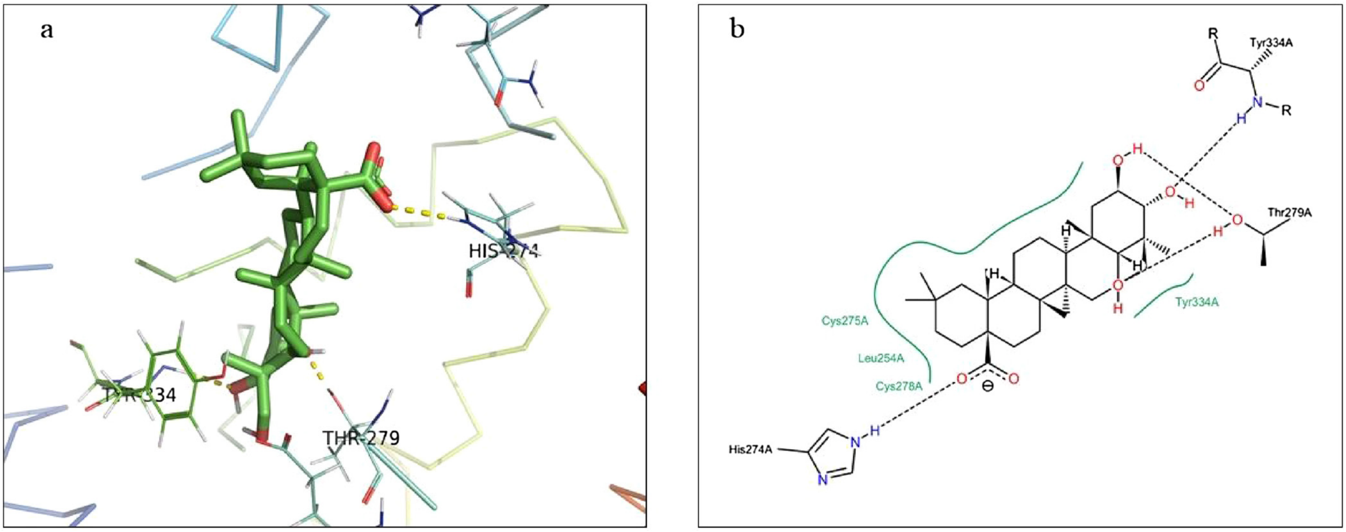

Fig. 2. (a) Docked and (b) hydrophobic orientations of AA with depiction of corresponding amino acid residue of PPARα active site.

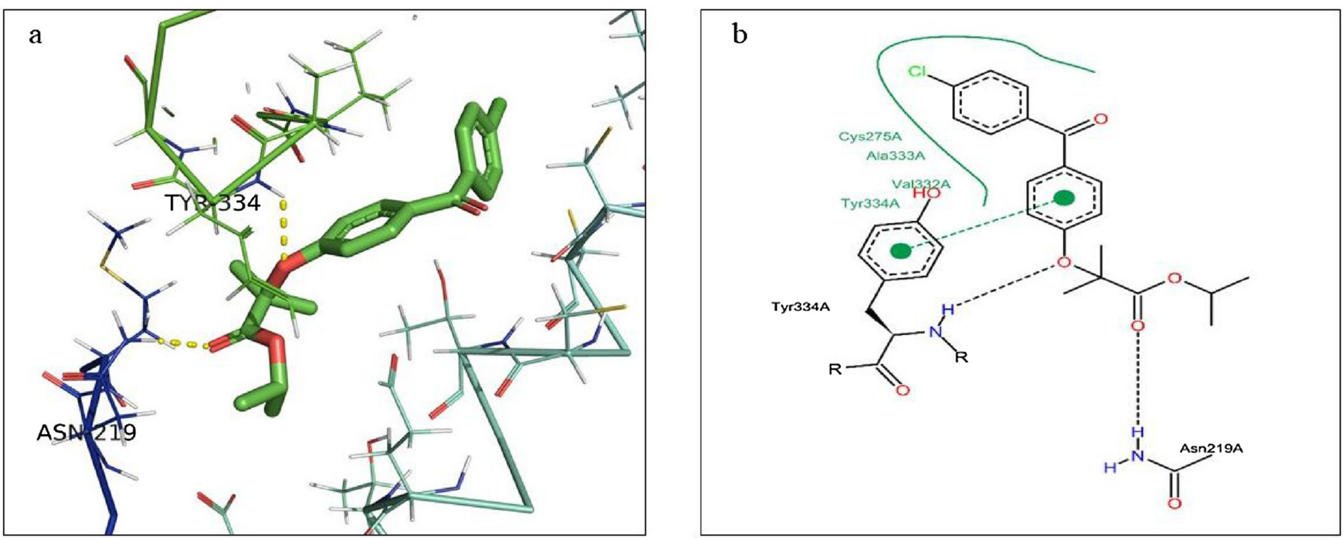

Fig. 3. (a) Docked and (b) hydrophobic orientations of fenofibrate with depiction of corresponding amino acid residue of PPARα active site.

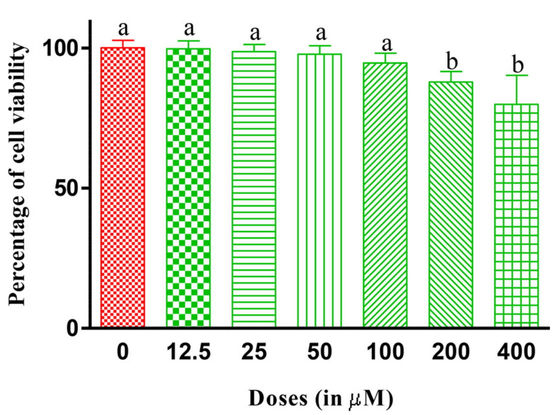

Fig. 4. Effect of AA on the viability of HepG2 cells after 24 h of treatment. Results are expressed as percent viability from normal, untreated cells; Values indicate mean ± SD of three independent experiments; Values having same alphabet in a graph did not deviate significantly from normal untreated cells and compounds at same dose level (Tukey's HSD; $P \leq 0.05$).

3.6. Effect of AA on the lipid and liver marker enzyme levels of HFD animals

NAFLD was successfully established after eight weeks of HFD feeding. In these animals, the serum lipid and enzymes levels of HFD fed animals were prominently elevated compared to the animals fed with NFD. The treatment with AA and fenofibrate significantly reduced the triglyceride levels by 71.61 and 66.47%, respectively. The treatment also improved the ratio of total cholesterol to HDL cholesterol in a significant manner (Table 4). The treatment with AA lowered the AST levels by 37.54 and 33.52% compared to the vehicle treated animals, at 25 and 50 mg/kg b.w. concentrations respectively in a highly significant manner ($P < 0.005$). It also lowered the ALT levels by 45.49 and 51.46% at 25 and 50 mg/kg b.w. concentrations respectively. The reductions in the AST and ALT levels at 50 mg/kg concentration were comparable with that of normal control. The treatment with AA reduced the levels of LDH in the plasma; this reduction was not statistically significant. At 50 mg/kg concentration, the treatment with AA significantly reduced the GGT level (Table 5).

Table 3
Effect of AA treatment on various parameters of lipid accumulation and lipotoxicity in PO-BSA induced steatotic HepG2 cells after 24 h of treatment.

Groups	% ORO	Triglyceride/Protein (mg/g)	% LDH release	ALT in the spent media	AST in the spent media
Normal control	100.0 ± 5.03[a]	21.31 ± 3.14[a]	4.51 ± 2.49[a]	33.09 ± 4.58[a]	18.89 ± 3.67[a]
Disease control	159.5 ± 17.25[b]	76.32 ± 4.98[b]	45.61 ± 4.70[b]	103.9 ± 6.65[b]	44.48 ± 8.52[b]
Fenofibrate (50 μM)	117.8 ± 13.64[a]	30.76 ± 1.97[a,c]	10.19 ± 3.20[a]	47.41 ± 4.71[c]	25.98 ± 2.23[a]
AA (12.5 μM)	119.8 ± 8.68[a]	33.83 ± 4.62[c]	16.77 ± 1.94[c]	50.26 ± 1.70[c]	30.31 ± 1.49[a]
AA (25 μM)	125.3 ± 5.25[a]	28.57 ± 3.86[a]	10.91 ± 0.70[a]	44.64 ± 1.18[a,c]	27.18 ± 2.95[a]
AA (50 μM)	102.1 ± 3.63[a]	25.67 ± 3.77[a]	3.65 ± 0.40[a]	40.40 ± 2.01[a,c]	23.00 ± 5.06[a]

Values indicate mean ± SD for three independent experiments; Values having same alphabets in a column did not vary significantly (One way ANOVA followed by Tukey's HSD; $P \leq 0.05$).

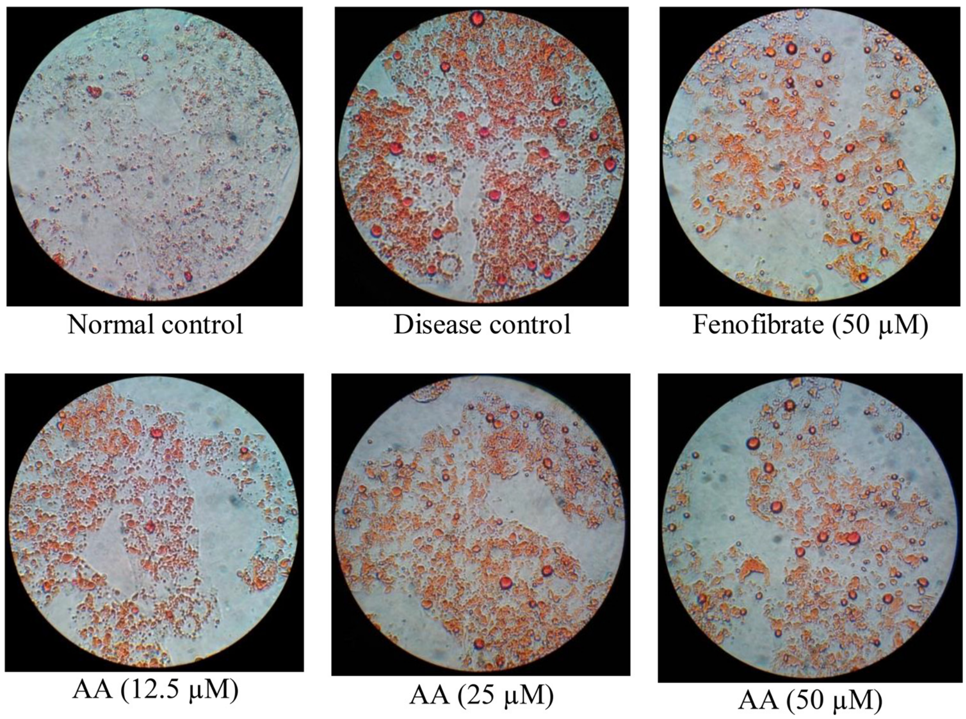

Fig. 5. Representative photomicrographs of PO-BSA induced steatotic HepG2 cells treated with AA.
The cells were treated with the compounds for 24 h and the images were taken after ORO staining at 400x magnification; the portions stained red indicated the presence of lipid accumulation.

Table 4
Effect of AA treatment on the serum lipid profile of HFD fed animals after four weeks of treatment.

Groups	Total cholesterol (mg/dL)	Triglycerides (mg/dL)	HDL cholesterol (mg/dL)	LDL cholesterol (mg/dL)	Total cholesterol/H L cholesterol
Normal control	55.61 ± 3.54[a]	55.96 ± 10.60[a]	25.10 ± 2.57[a]	19.32 ± 2.80[a]	2.23 ± 0.22[a]
Disease control	147.60 ± 11.93[b]	378.80 ± 43.75[b]	18.84 ± 3.30[a]	52.97 ± 8.03[b]	7.95 ± 0.90[b]
Fenofibrate	64.77 ± 9.13[a]	127.00 ± 26.32[c,d]	23.83 ± 4.60[a]	15.54 ± 5.49[a]	2.79 ± 0.60[a]
AA (25 mg/kg)	67.27 ± 7.39[a]	155.80 ± 26.90[c]	22.33 ± 4.53[a]	13.78 ± 2.83[a]	3.10 ± 0.55[a]
AA (50 mg/kg)	54.49 ± 5.93[a]	107.40 ± 21.07[d]	20.47 ± 3.47[a]	12.55 ± 5.74[a]	2.68 ± 0.20[a]

Values indicate mean ± Standard deviation of six animals; (One way ANOVA followed by Tukey's HSD; $P \leq 0.05$).

Table 5
Effect of AA treatment on the selected serum enzymes levels related to hepatotoxicity of HFD fed animals after four weeks of treatment.

Groups	AST (IU/L)[#]	ALT (IU/L)[#]	ALP (IU/L)[#]	LDH (IU/L)[$]	GGT (IU/L)[$]
Normal control	101.6 ± 15.7[a]	67.8 ± 11.5[a]	87.6 ± 9.1[a]	387.2 ± 59.0[a]	27.6 ± 5.8[a]
Disease control	171.8 ± 15.2[b]	187.5 ± 11.1[b]	205.2 ± 10.3[b]	1500.0 ± 469.5[b]	61.4 ± 15.1[b]
Fenofibrate (50 mg/kg)	114.0 ± 17.3[a]	101.1 ± 8.2[c]	113.6 ± 9.5[c,d]	557.8 ± 194.8[a]	29.8 ± 5.1[a,b]
AA (25 mg/kg)	107.3 ± 16.9[a]	102.2 ± 24.5[c]	125.9 ± 3.6[d]	672.6 ± 117.9[a,b]	31.8 ± 7.3[a,b]
AA (50 mg/kg)	114.2 ± 14.7[a]	91.0 ± 8.2[a,c]	101.9 ± 9.1[a,c]	575.4 ± 113.6[a,b]	18.2 ± 3.6[a]

Values indicate mean ± SD of six animals; Values having same alphabet in a column did not vary significantly ([#]One way ANOVA followed by Tukey's HSD; [$]Kruskal-Wallis test followed by Dunn's multiple comparison test; $P \leq 0.05$).

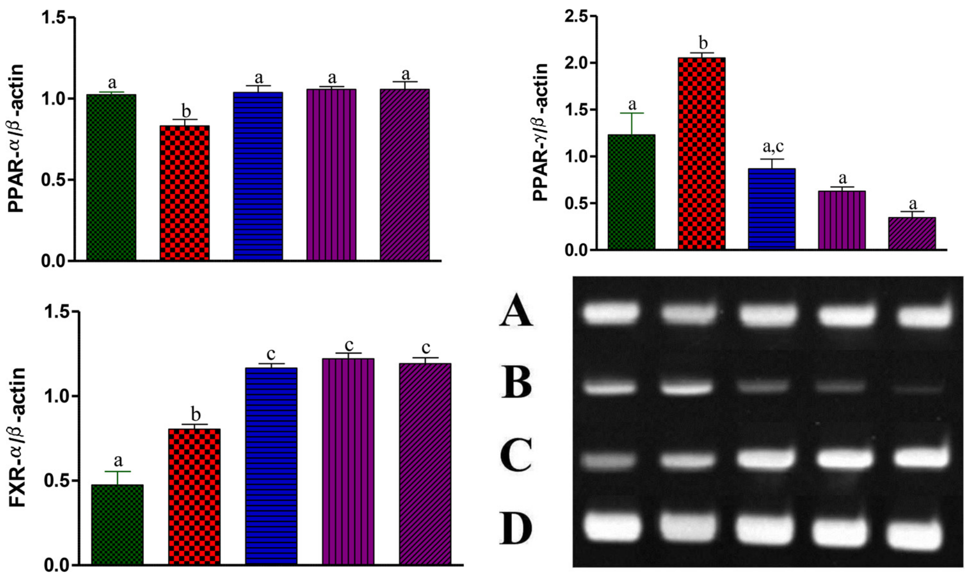

Fig. 6. Effect of AA on the expressions of hepatic genes of HFD fed rats after four weeks of treatment (A – PPARα; B – PPARγ; C – FXRα; and D – β-actin).
Gene expressions were normalised to the corresponding internal control (β-actin); Values indicate mean ± SD for three independent experiments; Normal control (lane 1), disease control (lane 2), fenofibrate (lane 3), AA (25 mg/kg) (lane 4) and AA (50 mg/kg) (lane 5) were compared; Values in graphs having same alphabets among them did not vary significantly (Tukey's HSD; $P \leq 0.05$).

3.7. Effect of AA on the hepatic gene expressions on HFD fed rats after four weeks of treatment

The administration of HFD to the animals downregulated PPARα expression and upregulated the expression of PPARγ and FXR mRNAs compared to NFD fed animals. Fenofibrate which was used as a PPARα agonist was found to reduce HFD induced steatosis through upregulation of PPARα. AA treatment to HFD fed animals upregulated the PPARα, and FXR mRNA expression to reduce steatosis and also downregulated PPARγ expression compared to untreated animals (Fig. 6). These reductions were statistically highly significant ($P < 0.005$).

3.8. Effect of AA on liver histopathology of HFD animals

The morphologies of the livers of HFD diet fed animals were pale and enlarged, while the animals treated with AA showed normal liver morphology. Histological examination of HFD diet fed animals showed multifocal fatty hepatocytes having non-eosinophilic cytoplasm and MNC infiltration. The treatment with fenofibrate and AA lowered these abnormalities (Fig. 7).

4. Discussion

NAFLD is one of the globally prevalent chronic liver diseases, predominantly associated with metabolic syndrome; it causes increased hepatic morbidity and mortality [38–41]. The current leading pharmacological interventions include weight reducing drugs, insulin sensitizers, lipid lowering agents, antioxidants, bile salts, mitochondrial fatty acid oxidation co-factors such as metformin, thiazolidinediones, pioglitazone, fibrates, orlistat and glucagon-like peptide-1. Some of these conventional prescriptions cause long term health risks such as cardiac failure, bone loss, pancreatitis, pancreatic and bladder cancer [42,43]. Lifestyle interventions such as diet modification, exercise, and weight reduction also decrease cardiovascular and diabetes risks along with the reduction of NAFLD [1,44]. Some preliminary researches have shown some positive effects of the natural products towards the management of NAFLD [45–48]. This study reports the curative effect of AA, one of the major compounds of *T. arjuna* on the models of NAFLD.

Preliminary *in silico* analysis showed that AA had no mutagenic, tumorigenic and irritant properties and molecular docking suggested that AA shared the same binding site of PPARα (TYR-334/HN) with fenofibrate; hence it can be a possible candidate drug for the treatment of NAFLD. *In vitro* cytotoxicity assessment of AA with HepG2 cells using MTT assay indicated that AA was not toxic up to 100 µM and had high GI_{25} value. The primary culture or immortalized hepatocyte cell lines such as HepG2, induced steatosis with palmitic and/or oleic acid is used as common model for screening the efficacy of compounds with anti-NAFLD effects [49]. Previous experiments showed that oleic acid treatment to HepG2 cells induced steatotic hepatocytes with morphological similarities to human steatotic hepatocytes developed by oleic acid accumulation [50]; but combination of palmitic and oleic acid produced comparatively high steatosis which mimics malignant chronic steatosis as found in humans [51]. During NAFLD progression there is an accumulation and disposal imbalance of Tg in the hepatic tissues which aggravates the lipotoxicity of hepatocytes. In this study, PO-BSA significantly elevated Tg concentration and lowered the viability of HepG2 cells. Treatment with AA significantly lowered Tg level and significantly increased the viability of HepG2 cells.

HFD fed rodent model is one of the predominantly employed animal models for studying NAFLD, since the features of this model significantly associated with metabolic syndrome, obesity and insulin resistance [52–54]. HFD is shown to cause direct and significant changes in the intracellular signaling pathways of hypothalamic target neurons. Though the exact pathogenetic mechanism still remains unclear [55], HFD fed animals mimic the various features of pathogenesis and histopathology as in human NAFLD [56] and thus this animal model was taken to assess the hepatoprotective effect of AA. This study showed that the treatment with AA significantly reduced the Tg and Tc as well as lowered the ratio of Tc to HDL-c. It indicated that the treatment with AA reduced the risk of NAFLD in the animals by reducing the lipid levels. Though the serum AST, ALT and ALP levels do not exactly reveal the hepatic injury, still they are used as important clinical markers of NAFLD progression [57,58]. In this study HFD elevated the serum transaminases and ALP levels indicating the NAFLD progression. The treatment with AA reduced these enzyme levels significantly indicating the curative effect of AA on NAFLD.

PPAR nuclear receptors are the members of the retinoid, steroid and thyroid hormone receptor superfamily of ligand-activated transcription factors [59,60]. PPARα, highly expressed in the hepatocytes, governs fatty acid transport and activates fatty acid oxidation thereby decreasing lipid storage [61–63]. PPARα knockout mice are susceptible to HFD with a significantly higher NAFLD scores [64]. Previous studies showed that HFD induced steatosis through the downregulation of PPARα and it presumably was mediated by increased circulating free fatty acid [65,66]. In this study also administration of HFD reduced PPARα expression and the treatment with AA and fenofibrate reversed

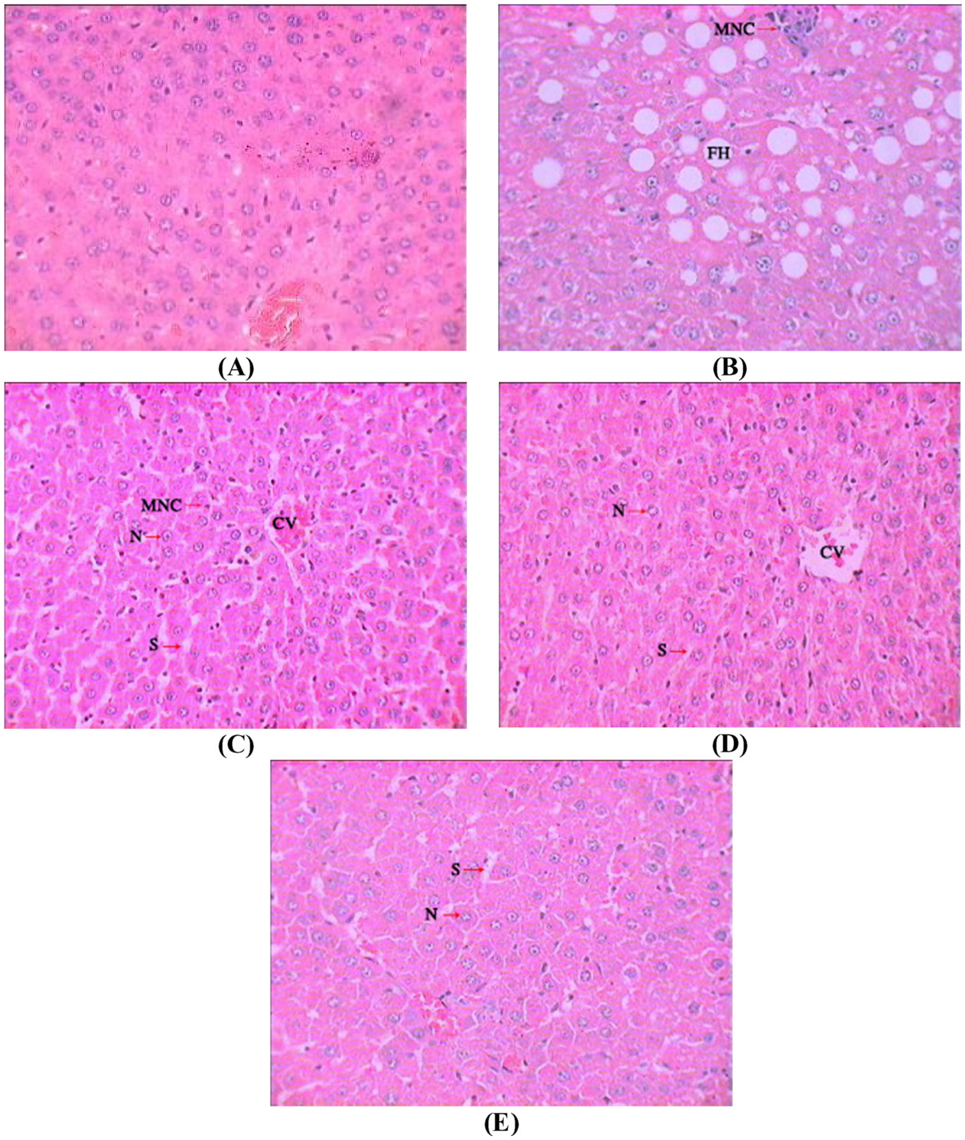

Fig. 7. Representative histological images of liver sections of HFD fed rats after four weeks of treatment with AA.
Livers were harvested, fixed in 10% buffered formalin and stained with haematoxylin – eosin. Images were taken at 400× magnification. Representative microphotographs were presented for (A) Normal control, (B) Disease control, (C) Fenofibrate (50 mg/kg), (D) AA (25 mg/kg), (E) and AA (50 mg/kg), respectively [CV – Central Vacuole; FH – Fatty hepatocytes; MNC – Mononuclear cells; N – Nucleus; S – Sinusoids).

this abnormality. Studies showed the agonistic effect of fibrates such as fenofibrate [67] and it was shown that the treatment with fenofibrate reduced hepatic steatosis and inflammation in HFD fed animals [68]. A recent study with AA showed that AA can bind with PPARα and stabilize the ligand binding domain of this ligand [69]. This study was also in concordance with the previous reports where the treatment with AA and fenofibrate significantly increased the PPARα expression compared to the HFD fed animals. This report also indicated that this might be the possible molecular pathway of AA to ameliorate NAFLD.

PPARγ majorly expresses in adipose tissue, where it regulates the adipocyte differentiation, adipogenesis and lipid metabolism, transcription process and glucose metabolism [70–72]. Previous clinical and preclinical reports showed an upregulation in the PPARγ expression in hepatocytes during NAFLD [73–75]. This study also showed that HFD feeding caused an upregulation of PPARγ and the treatment with AA significantly reduced this effect. This study gave an idea that AA treatment might selectively upregulate the expression of PPARα. FXR is a ligand-activated transcription factor, expressed in many tissues including liver and involved in lipid, glucose and energy homeostasis as well as in inflammation [76,77]. Feeding animals with HFD caused an upregulation of FXR [78] and treatment with FXR agonist prevented diet induced hepatic steatosis [79]. The results of histological analysis were in agreement with the biochemical and molecular findings. In this study, the treatment with AA significantly increased the PPARα and FXR gene expression and decreased PPARγ gene expression indicating that it may be taken as a lead molecule to identify some novel non-steroidal, anti-NAFLD agents.

5. Conclusion

This study provided preliminary evidence on the beneficial effect of AA for the treatment of NAFLD and this study had some limitations. This study documented only the short term efficacy of AA; dose dependency of AA was not assessed thoroughly and the molecular mode of action was assessed only at mRNA level with few selected genes. Detailed investigations on the long term efficacy of AA on the progressive stages of NAFLD and its exact molecular mode of action may yield some useful leads for the better management of NAFLD.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:https://doi.org/10.1016/j.biopha.2018.08.019.

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